

Carry-select sketch

Ripple carry waits for each stage before the next begins

A5 plus 3C starter

- Starter: eight-bit add, A equals hex A5, B equals hex three C
- Lower nibble: five plus C gives sum one and C4 equals one
- Upper dual paths: Cin zero gives D, Cin one gives E
- Mux selects the Cin-one path because C4 is one
- Final sum is hex E1, Cout zero
- Step through phases

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Carry-select adder sketch

Upper nibble runs **two** adders in parallel (Cin=0 and Cin=1); a mux picks the right sum when the lower carry arrives.

Starter example: A=0xA5, B=0x3C — animate dual upper paths, then mux on carry from the lower nibble.

Load starter example

Challenges (1/22)

Quiz: CSA. Adder that muxes dual Cin paths is a? Answer: carry-select

Answer

Show hint

Check

Next

Idle

quiz-csa

quiz-mux

quiz-speculate

quiz-c4

starter

example

compute

phase

run

sum-e1

c4-zero

c4-one

overflow

demo

explain

dual-visible

mux-phase

code-mux

paths-differ

reset

pick-match

full-path

Core ideas

Speculate

Upper block adds twice — Cin 0 and 1 — in parallel.

Select

Real Cin (from lower) picks the correct sum via mux.

Why

Hide upper carry chain latency behind lower compute.

Carry-select adder starter

learn_digital — carry select adder

Workbook practice

- On paper, sketch an eight-bit carry-select built from two four-bit blocks
- For A_5 plus three C , compute lower sum and C_4 , both upper speculative sums
- Tabulate C_{in} zero versus C_{in} one paths
- Name the trade-off

Pitfalls to watch

- Do not confuse speculation with the final answer, the mux must wait for real C4
- Both upper paths are computed in parallel; only one is selected
- C4 is carry out of the lower nibble, not the full eight-bit Cout
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run all phases on the starter and confirm sum E1
- On paper, draw the dual upper paths and mux
- When you are ready, take the short quiz, then continue to Booth encode